Cover Letter for GSI Position (EAS 531)

jiayuanj@umich.edu Jiayuan Jiang

April 6, 2026

Prof. Kim Diver
School for Environment and Sustainability
University of Michigan, Ann Arbor

Dear Prof. Diver,

I am writing to express my strong interest in the GSI position for EAS 531: Principles of GIS. I am currently a Master's student in Geospatial Data Science at the University of Michigan, and my long-term goal is to pursue research in GeoAI, focusing on the integration of spatial data and machine learning for understanding complex geographic processes.

I completed my undergraduate degree in Geographic Information Science at Huazhong Agricultural University, where I received comprehensive training in GIS and remote sensing. My coursework includes Principles of GIS, GPS Technology, Introduction to Remote Sensing, Hyperspectral Remote Sensing, Quantitative Remote Sensing, Active Remote Sensing, UAV Remote Sensing, and GIS Development. This background has provided me with a solid theoretical foundation and strong practical skills in geospatial analysis.

I have extensive experience using ArcGIS Pro and am highly proficient in ArcPy for spatial data processing, including batch raster operations, data preprocessing, and geoprocessing workflows. Through both coursework and research, I have developed strong debugging skills and am familiar with common issues in GIS workflows. I also have experience with QGIS, including plugin-level development, which allows me to understand GIS tools beyond surface-level usage.

I completed EAS 531 in the previous semester and received an A+ grade. This experience allows me to understand the course structure, lab workflow, and common challenges students encounter. I am confident in my ability to guide students through lab exercises, assist with troubleshooting, and provide clear explanations of both concepts and technical steps.

In addition, I am currently working as a Research Assistant under Professor Kai Zhu on a GeoAI project focused on forest disturbance mapping using multi-temporal satellite imagery and machine learning. This experience has strengthened my ability to work with large-scale geospatial datasets, develop analytical workflows, and apply computational methods to real-world environmental problems.

I am particularly interested in this position because I enjoy helping others solve technical problems and learn complex tools. I am comfortable supporting students during lab sessions, assisting with debugging in real time, and ensuring they can complete assignments effectively. I believe my technical background, familiarity with the course, and interest in teaching would allow me to contribute positively to the learning environment.

Thank you for your time and consideration.

Sincerely,

Jiayuan Jiang

Last updated in February 2026

Jiayuan Jiang

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resume-jiayuan-jiang.vercel.app | github.com/jiayuan-jiang

Education

University of Michigan, Ann Arbor Aug 2025 – Present

Master of Science in Geospatial Data Science

School for Environment and Sustainability (SEAS)

GPA: 4.0 / 4.0

Huazhong Agricultural University Sep 2021 – Jun 2025

Bachelor of Science in Geographic Information Science

College of Resources and Environment, Wuhan, China

GPA: 86.3 / 100

Technical Skills

Programming: Python, R, JavaScript, SQL, Bash

Machine Learning: PyTorch, Model Distillation, Vision-Language Models (CLIP), Embedding-Based Retrieval

System Development: RESTful APIs, Distributed Inference Pipelines, Database Design, Cloud Infrastructure

Geospatial Computing: GeoPandas, GDAL, Rasterio, ArcGIS Pro, Google Earth Engine

Programs

PIVOT — Participatory GIS Urban deVelOpment Tool Feb 2026 – Present

https://pivot-model.com/

Web Infrastructure & Full-Stack Developer | Research Assistant, University of Michigan (GLISA)

Led full-stack development for a NOAA-funded participatory WebGIS platform supporting climate adaptation and urban redevelopment planning.

Implemented new platform features including client-side draft persistence with local caching and draft upload workflows that allow users to submit planning scenarios for automated analysis and summarization.

Managed AWS-based cloud infrastructure, database operations, and production deployment to ensure stable and secure access for planners and stakeholders across 18 pilot cities.

Nature Observer — Plant Identification and Learning Platform Oct 2025 – Present

https://nature-observer.vercel.app/

Backend & Algorithm Developer | Team Project in collaboration with Matthaei Botanical Gardens

Developed a domain-adapted plant recognition model via transfer learning on a curated botanical dataset from Matthaei Botanical Gardens, improving classification accuracy for species present within the garden environment.

Evaluated open-source vision architectures and applied model distillation to compress parameters for lightweight client-side inference via TensorFlow.js, enabling low-latency execution without dedicated backend infrastructure.

Built a CLIP-based multimodal tagging pipeline to automatically generate fine-grained visual annotations for large-scale plant images from iNaturalist, creating a high-precision visual tagging system that enables accurate image grounding for educational content generation.

Designed an LLM-driven application architecture using schema-constrained generation to produce deterministic JSON outputs, enabling a programmatic pipeline that converts model reasoning into structured quizzes and interactive visual learning flows.

Fire Spread Simulation Platform Summer 2024

https://project-wildfire.vercel.app/

Full-Stack Developer | Summer Intern, Zongdi Cyber Group Co., Ltd

Implemented a cellular automata wildfire propagation model across raster grids, incorporating wind-driven transition rules and terrain-dependent spread probabilities.

Integrated real-time weather inputs including wind speed and direction with terrain and land-surface datasets to parameterize fire spread probabilities and simulate evolving fire fronts.

Built a full-stack WebGIS platform using Vue3 and Leaflet for interactive visualization and a Flask backend for simulation computation, enabling users to initialize ignition points and explore predicted fire propagation scenarios.

Research Experience

Spatial Heterogeneous Graph Neural Network Optimization May 2024 – Jul 2025

Research Assistant | Beijing Digsur Science and Technology Co., Ltd

Proposed a novel heterogeneous graph neural network framework that integrates global and local spatial clustering with multi-scale feature fusion to enhance spatial representation learning in heterogeneous urban graphs.

Designed and implemented the architecture in PyTorch, combining spatially constrained clustering with Skater and Ward methods, local Moran's I heterogeneity detection, and attention-gated multi-head fusion for adaptive spatial feature aggregation.

Achieved a 22.7% reduction in RMSE and an 18.3% reduction in MAE compared to baseline GCN and GBK models, demonstrating improved generalization and interpretability in urban geospatial prediction; drafted and filed a patent based on this framework.

Interaction Between Riparian Vegetation and Reservoir Evaporation Feb 2023 – Jun 2024

Research Assistant | Huazhong Agricultural University

Conducted research on the impact of the Three Gorges Reservoir on vegetation evapotranspiration under drought stress as part of a National Undergraduate Innovation and Entrepreneurship Project.

Performed large-scale preprocessing of remote-sensing and meteorological datasets from 1998 to 2022 using ArcPy for batch resampling, clipping, and alignment of raster layers to ensure consistency across inputs.

Improved and validated a Surface Energy Balance System (SEBS) evapotranspiration model from open-source code, analyzing evapotranspiration gradients within reservoir buffer zones through zonal statistics, Kriging interpolation, and correlation analysis.

Computed NDVI indices using the Google Earth Engine cloud platform and integrated drought indicators to quantify vegetation response, revealing that reservoir effects mitigated excessive evapotranspiration within 200 meters of the waterline.

華中農業大學
HUAZHONG AGRICULTURAL UNIVERSITY

本科学生成绩单
Academic Transcript for Undergraduate Student

Name: Jiang Jiayuan Student number:2021303220208 Gender:Male Date of Birth:15/04/2003 Date of Admission:09/2021

College: College of Resources and Environment Major: Geographical Information Science Date of Graduation:06/2025

2021-2022 First semester

Course	Credits	Score
Physical Geography	2.0	80
Practice for Physical Geography	1.0	82
Experiment of Physical Geography	1.0	88
Cartography	2.0	80
Cartography Experiment	1.0	88
Seminar on Ecology and Environment Monitoring	1.0	89
Foundation for Engineering	1.0	90
Calculus C	5.0	84
Writing and Communication	1.0	86
College English Integrated Course (Advanced)	2.0	82
Tennis B	1.0	82
Outline of Modern Chinese History	3.0	80
Innovation and Entrepreneurship Foundations for Undergraduates	1.0	97#
Military theory	2.0	74

2021-2022 Second semester

Course	Credits	Score
Psychological Development and Guidance of College Students	1.0	92
The Practice of Mental Health Promotion for Undergraduates	1.0	Pass#
Principle of Geographical InformationSystem Experiment A	1.0	85
Principle of Geographical Information System A	2.0	78
Linear Algebra B	2.0	98
College Physics B	4.0	84
College Physics Experiment	1.0	75
College English: Viewing,Listening and Speaking (Advanced)	2.0	86
Tennis A	1.0	81
History of the PRC Research	1.0	90
Ideology Morality and Rule of Law	3.0	82
Situation and Policy Education (1)	1.0	82
Software Engineering B	2.0	82
C++ Programming	2.0	97
Military skills	2.0	90

2022-2023 First semester

Course	Credits	Score
Digital Terrain Models	1.0	80
Introduction to Remote Sensing A	2.0	82
Digital Terrain Models Experiment	1.0	91
Probability and Statistics A	4.0	66
IELTS (International English Language Testing System) Speaking	2.0	93
Public French	2.0	82
Badminton C	1.0	79
Basic Theory of Mao Zedong Thoughts and Socialism with Chinese Characteristics	3.0	92
Basic Theory of Xi Jinping Thought on Socialism with Chinese Characteristics for a New Era	3.0	86
General Surveying Practice	3.0	83
Surveying Practice A Experiment	1.0	92
Data Structure B	3.0	94

2022-2023 Second semester

Course	Credits	Score
Introduction to Agronomy	2.0	82
Global Position System Principles and Applications	1.0	85
Artificial Intelligence and Machine Learning	2.0	85
Global Position System Principles and Applications Experiment	1.0	88
Digital Image Processing Experiment	1.0	86
Comprehensive experiment of remote sensing	1.0	89
The Research of the Campus Land Use Changes on Temporal and Spatial Based on "3S" technology	1.0	89#
Principle of Spatial Database	3.0	86
Principle of Spatial Database Experiments	1.0	93
Qualitative Remote Sensing	2.0	84
Digital Image Processing	2.0	89
Tzu chih t'ung chien and its theory of management	1.0	91
Mathematical Modeling B	2.0	91
English for General Academic Purposes (Agriculture Science 1)	2.0	95
Outward Bound C	1.0	91
Basic Knowledge of Marxism	3.0	84
Social Practice	1.0	90

2023-2024 First semester

Course	Credits	Score
Specialty English of GIS	1.0	89
Hyperspectral Remote Sensing	2.0	95
Soil and fertilizer science B	3.0	83
Resource Survey and Assessment B	2.0	84
Resource Inventory and Assessment Practice	2.0	89
Landscape Ecology Experiment	1.0	92
Low altitude remote sensing application on agriculture and forest	1.0	97
Landscape Ecology	2.0	85
Computer Networks B	3.0	90
Python Programming	2.0	94

2023-2024 Second semester

Course	Credits	Score
Geographic Information Processing and Modeling	1.0	88
Quantitative geography	1.0	74
Application of Remote Sensing Practice	2.0	93
Geographic Information Processing and Modeling Experiment	2.0	92

Note: ①*: retake courses; ②#: Not counted in GPA; ③△: Identified Courses. Number:ZZDY HGBN HNJM ACGJ AMPD BKIB NBBA MNGL TU

教务处长签章 Dean 教务处（盖章） Office of Undergraduate Academic Affairs 打印日期 Issue Date June 22, 2025

華中農業大學
HUAZHONG AGRICULTURAL UNIVERSITY

本科学生成绩单
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Name: Jiang Jiayuan Student number:2021303220208 Gender:Male Date of Birth:15/04/2003 Date of Admission:09/2021

College: College of Resources and Environment Major: Geographical Information Science Date of Graduation:06/2025

Geographic information Systems Design and Development Experiment	1.0	A
Quantitative geography Experiment	1.0	90
Active Remote Sensing Experiment	1.0	93
Uncertainties in Spatial Data Processing	2.0	86
Active Remote Sensing	1.0	91
Geographic Information Systems Design and Development	2.0	87
The Experiment of Big Data Application in Agriculture	1.0	91
Appreciation of Traditional Chinese Painting	2.0	95
Appreciation of Classical Western Literature	2.0	95
2024-2025 First semester		
Course	**Credits**	**Score**
Spatial? data? analysis? and? global? environmental? changing	2.0	92
Comprehensive Practice	2.0	93
Land Use Planning B	2.0	75
2024-2025 Second semester		
Course	**Credits**	**Score**
Volunteer Service	1.0	Pass#
Aesthetic Practice	1.0	Pass#
The Art of Living Today——Garden, Incense and Tea	2.0	99#
Graduation Thesis	6.0	87
Graduation Internship	2.0	82
Situation and Policy Education (2)	1.0	86
Weighted average score	86.32	

Note: ①*: retake courses; ②#: Not counted in GPA; ③△: Identified Courses; Number:ZZDY BKHG BGFI FLGB FKJD EAFE KEEL BKGJ TU

教务处长签章 Dean 教务处（盖章） Office of Undergraduate Academic Affairs 打印日期 Issue Date June 22, 2025

華中農業大學
HUAZHONG AGRICULTURAL UNIVERSITY

本科学生成绩单
Academic Transcript for Undergraduate Student

姓名：江佳圆　学号：2021303220208　性别：男　出生日期：2003-04-15　入学日期：2021-09

学院：资源与环境学院　专业：地理信息科学　毕业日期：2025-06

2021-2022 学年第 1 学期		
课程名称	学分	成绩
自然地理学	2.0	80
自然地理学综合实习	1.0	82
自然地理学实验	1.0	88
地图学	2.0	80
地图学实验	1.0	88
地理大数据在生态环境监测中的应用	1.0	89
工程基础	1.0	90
微积分C	5.0	84
写作与沟通	1.0	86
大学英语综合（高级）	2.0	82
网球B	1.0	82
中国近现代史纲要	3.0	80
大学生创新创业基础	1.0	97#
军事理论	2.0	74

2021-2022 学年第 2 学期		
课程名称	学分	成绩
大学生心理发展与指导	1.0	92
大学生心理素质拓展	1.0	合格#
地理信息系统原理A实验	1.0	85
地理信息系统原理A	2.0	78
线性代数B	2.0	98
大学物理B	4.0	84
大学物理实验	1.0	75
大学英语视听说（高级）	2.0	86
网球A	1.0	81
新中国史研究	1.0	90
思想道德与法治	3.0	82
形势与政策（一）	1.0	82
软件工程B	2.0	82
C++语言程序设计	2.0	97
军事技能	2.0	90

2022-2023 学年第 1 学期		
课程名称	学分	成绩
数字地面模型	1.0	80
遥感导论A	2.0	82
数字地面模型实验	1.0	91
概率论与数理统计A	4.0	66
雅思口语	2.0	93
公共法语	2.0	82
羽毛球C	1.0	79
毛泽东思想和中国特色社会主义理论体系概论	3.0	92
习近平新时代中国特色社会主义思想概论	3.0	86
普通测量学	3.0	83
测量学A实验	1.0	92
数据结构B	3.0	94

2022-2023 学年第 2 学期		
课程名称	学分	成绩
农学概论	2.0	82
全球定位原理与应用	1.0	85
人工智能与机器学习	2.0	85
全球定位原理与应用实验	1.0	88
数字图像处理实验	1.0	86
遥感综合实验	1.0	89
基于3S的土地利用时空变迁研究	1.0	89#
空间数据库原理	3.0	86
空间数据库原理实验	1.0	93
定量遥感	2.0	84
数字图像处理	2.0	89
资治通鉴与管理之道	1.0	91
数学建模B	2.0	91
通用学术英语（农林1）	2.0	95
拓展运动C	1.0	91
马克思主义基本原理	3.0	84
社会实践	1.0	90

2023-2024 学年第 1 学期		
课程名称	学分	成绩
GIS专业英语	1.0	89
高光谱遥感	2.0	95
土壤肥料学B	3.0	83
资源调查与评价B	2.0	84
资源调查与评价劳动实习	2.0	89
景观生态学实验	1.0	92
农林无人机遥感应用	1.0	97
景观生态学	2.0	85
计算机网络B	3.0	90
Python程序设计	2.0	94

2023-2024 学年第 2 学期		
课程名称	学分	成绩
GIS空间分析	1.0	88
计量地理学	1.0	74
遥感应用综合实习	2.0	93
GIS空间分析实验	2.0	92
地理信息系统设计与开发综合实验	1.0	A
计量地理学实验	1.0	90

备注:①*表示重修；②#表示不计入GPA课程；③△表示认定课程　　\KD ANMA DAAG BDKL TZZD YEAH GEGD FCHE FHAK D

教务处长签章 Dean　　教务处（盖章） Office of Undergraduate Academic Affairs　　打印日期 Issue Date 2025-06-22

華中農業大學
HUAZHONG AGRICULTURAL UNIVERSITY

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主动式遥感实验	1.0	93
空间数据的不确定性分析	2.0	86
主动式遥感	1.0	91
地理信息系统设计与开发	2.0	87
农业大数据处理与应用实验	1.0	91
中国传统绘画鉴赏	2.0	95
西方文学经典赏析	2.0	95
2024-2025 学年第 1 学期		
课程名称	学分	成绩
空间数据分析和全球环境变化	2.0	92
专业综合实习	2.0	93
土地利用规划B	2.0	75
2024-2025 学年第 2 学期		
课程名称	学分	成绩
志愿服务	1.0	合格#
美育实践	1.0	合格#
现代生活美学——花香茶之道（MOOC）	2.0	99#
毕业论文	6.0	87
毕业实习	2.0	82
形势与政策（二）	1.0	86
加权平均分	86.32	

备注:①*表示重修；②#表示不计入GPA课程；③△表示认定课程 AQD BKIB NBBA MNGL TZZD YJIC AJAK NKBI NNAQ D

教务处长签章 Dean 彭小川

教务处（盖章） Office of Undergraduate Academic Affairs

打印日期 Issue Date 2025-06-22

华中农业大学本科学生成绩单相关说明

Official Explanations on the Grading System for Undergraduate Student's Academic Transcript, HuaZhong Agricultural University

一、平均学分绩点计算公式(GPA Formula)

百分制成绩 (Hundred-mark system)	成绩等级 (Grade)		学分绩点 (Grade points)
90~100	优 Excellent	A	4
85~89		A-	3.7
82~84		B+	3.3
78~81	良 Good	B	3
75~77		B-	2.7
72~74		C+	2.3
68~71	中 Satisfactory	C	2
64~67		C-	1.5
60~63	及格 Passing	D	1
60以下	不及格 Fail	F	0

$$\text{平均学分绩点} = \frac{\Sigma\,(\text{课程学分} \times \text{所获学分绩点})}{\Sigma\,\text{课程学分}}$$

$$\text{Grade Points Average (GPA)} = \frac{\Sigma\,(\text{course credits} \times \text{grade points of the course})}{\Sigma\,\text{course credits}}$$

二、学时、学分标准(Hours of Instruction and Credits)

1学时=45分钟；理论课程16学时=1学分；实验课程30学时=1学分；实践课程2周=1学分。

One class hour is 45 minutes. A student will receive one credit for satisfactorily completing sixteen (16) hours of instruction, and one credit for thirty (30) hours of laboratory work, and one credit for completing two weeks of practice course.

三、毕业日期(Graduation Date)

成绩单中的毕业日期为学生预计毕业时间，如取得毕业证则为实际毕业时间。

The date on the academic transcript is the anticipated graduation date, and the actual graduation date is indicated on the student's diploma.

湖北省武汉市洪山区狮子山街1号　邮编：430070
电话：027-87281542　传真：027-87282023
No.1, Shizishan Street, Hongshan District, Wuhan, Hubei Province, 430070, P.R.China
Tel:86-27-87281542　Fax:86-27-87282023

THE UNIVERSITY OF MICHIGAN - ANN ARBOR

Unofficial Transcript - Not an Official Transcript

Jiang,Jiayuan
UM ID: 46287929
Uniqname: JIAYUANJ

Date: Mar 7, 2026

Jiang,Jiayuan (734) 741-3820
1717 Broadview Str. Apt. 214
Ann Arbor, MI 48105-1845
United States

Fall 2025		**Rackham**	**Grade**	**Hours**	**MSH**	**CTP**	**MHP**
EAS	506	Intro to R	S	1.00	0.00	1.00	0.00
EAS	510	Sci&Prac of Soc Chng	A	3.00	3.00	3.00	12.00
EAS	531	Principles of GIS	A+	4.00	4.00	4.00	17.20
EAS	548	Intro Geovisual	A	1.50	1.50	1.50	6.00
EAS	648	Adv Geovisual	A	1.50	1.50	1.50	6.00
SI	544	Intr Stat&Data Analy	A+	3.00	3.00	3.00	12.90
Term Total		**GPA: 4.000**		**14.00**	**13.00**	**14.00**	**54.10**
Cumulative Total		**GPA: 4.000**			**13.00**	**14.00**	**54.10**

Winter 2026		**Rackham**	**Grade**	**Hours**	**MSH**	**CTP**	**MHP**
Elections as of: 03/07/2026							
EAS	700	Master's Thesis		5.00	0.00	0.00	0.00
EECS	484	Database Mgt Syst		4.00	0.00	0.00	0.00
EECS	504	Fnd Computer Vision		3.00	0.00	0.00	0.00
EECS	545	Machine Learn (CSE)		3.00	0.00	0.00	0.00
ENGR	555	Grad Multidisc Proj		2.00	0.00	0.00	0.00
		SIM Research Team					
SI	630	NLP Algorithm People		3.00	0.00	0.00	0.00
Elected Term Hours				**20.00**			

Academic Statistics for Rackham		**MSH**	**CTP**	**MHP**
Total to Date	**GPA: 4.000**	**13.00**	**14.00**	**54.10**

Program Action History: **Environ & Sustainability MS**

02/27/2025 Matriculation
Environ & Sustainability MS / Geospatial Data Sciences

Academic Previous Experience

Huazhong Agricultural University	42, China
Bachelor of Science	06/01/2025

End of Unofficial Transcript